RECAST: Methodology Supplement

This supplement provides methodological details and derivations for RECAST, including its architecture, constrained optimization, training objectives, gradient replay, and computational complexity.

S1. Notation and Operator Conventions

A. Decision-time state

Let t index trading decisions, $i \in \{1, \dots, N_t\}$ index the aligned stock axis, and k index within-decision refinement rounds. We suppress t when describing one decision. The mask $v_i \in \{0, 1\}$ denotes an available stock representation, and $\mathcal{V} = \{i : v_i = 1\}$. An observation is usable only if available at the decision timestamp. We use the Center for Research in Security Prices (CRSP) permanent security identifier, PERMNO, to align a stock across dates and ticker changes [1].

The fused stock representation is $h_i \in \mathbb{R}^d$, with $d = 256$. The initial and refined relation states are $r_i^0, r_i^k \in \mathbb{R}^d$. The holdings-conditioned state $\hat{r}_i^k$ includes a contribution prediction. A signed allocation w_i^k is positive for a long position and negative for a short position, expressed as a fraction of portfolio net asset value. The latest standardized stock attributes and their observation mask are $x_i, b_i \in \mathbb{R}^S$, where $S = 71$ and $b_{if} \in \{0, 1\}$. These latest attributes are distinct from the complete daily input window.

There are $J = 32$ diagnostic requests. Request j consists of a state $\psi_j \in \mathbb{R}^d$, retrieval key $q_j \in \mathbb{R}^{32}$, nonnegative source distribution $u_j \in \mathbb{R}^N$, capital limit $m_j \geq 0$, and side $\sigma_j \in \{+1, -1\}$. A transfer plan p belongs to request $j(p)$, has target distribution π_p, signed direction d_p, predicted gain g_p, and factor vector $f_p \in \mathbb{R}^8$. We write $D = [d_1, \dots, d_P] \in \mathbb{R}^{N \times P}$, $F = [f_1, \dots, f_P]^\top \in \mathbb{R}^{P \times 8}$, and $g = (g_1, \dots, g_P)^\top$. The request-membership matrix is $(T_s)_{jp} = \mathbf{1}\{j(p) = j\}$. We distinguish the internal revision-penalty coefficient λ_{rep} from the base execution-fee rate c_f. Both are fixed at 0.0005 in the implemented configuration; equality of their numerical values does not identify their optimization and accounting roles.

B. Numerical and neural operators

Concatenation is denoted by $\|$. A named two-layer map $f_{a \to b \to c}$ applies an affine map, the Gaussian error linear unit (GELU) introduced by Hendrycks and Gimpel, and a second affine map with the indicated widths. For a scalar input x, $\mathrm{GELU}(x) = x\Phi(x)$, where Φ is the standard normal cumulative distribution function [2, App. B.3]. Differently named maps have separate parameters. Parameters are shared across stocks and decisions, and refinement operators are shared across rounds. The normalization $\mathrm{unit}(z) = z/\max(\|z\|_2, 10^{-12})$ operates on one vector. Unless stated otherwise, denominator stabilization uses $\varepsilon = 10^{-8}$. The operator sg keeps the forward value and stops its gradient, as in PyTorch's detach operation [3]. For logits z, the Gibbs normalization [4] restricted to a nonempty support $\mathcal{S}$ gives

$$[\mathrm{softmax}_{\mathcal{S}}(z)]_i = \frac{\mathbf{1}\{i \in \mathcal{S}\} \exp(z_i)}{\sum_{\ell \in \mathcal{S}} \exp(z_\ell)}. \tag{S1}$$

An empty support produces the zero vector, retaining its unallocated budget as cash. Invalid entries never enter a masked normalization. Observed-feature averages divide by observed mass; a stored zero for a missing feature is not interpreted as an observed economic value.

S2. Dual-Frequency Encoding and Shared Relational Summaries

A. Sequence construction and temporal encoders

The daily input contains 252 observations with 71 channels. These comprise 29 base variables, 20 trend and liquidity extensions, and 22 industry-relative, asymmetric-response, tail, and intraday-shape variables. For example, the 21- and 63-day illiquidity features average the ratio of absolute daily return to dollar volume, following Amihud's measure [5]. The monthly input contains 24 observations with 51 channels, partitioned into two consecutive 12-month blocks. Both blocks share one monthly encoder. The daily and monthly encoders have separate parameters. Monthly channels provide longer-horizon firm and stock context, including size and valuation, profitability and investment, and historical return, risk, and trading-friction characteristics. Monthly observations end at the latest completed month available before the decision. Daily channels follow the recorded feature-availability timestamp rather than their eventual file-publication order.

The per-stock sequence encoder follows AlphaPortfolio's temporal representation module [6]. Its sinusoidal positions, scaled dot-product attention and residual blocks follow the Transformer [7]. Each encoder projects features to $d = 256$, adds sinusoidal positions, and uses one Transformer layer with four heads, feed-forward width 1024, dropout 0.1, and residual post-normalization. We use Ba, Kiros, and Hinton's layer normalization in the form without learned gain or bias studied by Xu et al. [8]. We apply dropout [9] during training. The input projection has no bias. For one temporal head, with $d_h = 64$,

$$\begin{aligned} \mathrm{Attn}(Q, K, V) &= \mathrm{softmax}(QK^\top/\sqrt{d_h})V, \\ H' &= \mathrm{LN}\{H + \mathrm{Drop}(\mathrm{MHA}(H))\}. \end{aligned} \tag{S2}$$

The feed-forward sublayer applies $Z = \mathrm{Drop}_{\text{in}}(\mathrm{ReLU}(W_1 H' + b_1))$, followed by $\mathrm{LN}\{H' + \mathrm{Drop}_{\text{out}}(W_2 Z + b_2)\}$. Attention probabilities also use dropout during training. The last temporal position supplies

the stock-level state. Learned gates regulate information flow in Highway Networks [10]. Our fusion gates the monthly correction to the daily state and scales it by the observed monthly coverage.

Let δ_i^D contain daily channel-missing fractions, $c_{ib} \in [0,1]$ the usable monthly coverage of block $b \in \{1,2\}$, and h_{ib}^M its encoded state. The fusion is

$$\begin{aligned} h_i^D &= \mathrm{Enc}_D(X_i^D) + W_{\mathrm{miss}}\delta_i^D, \\ h_i^M &= \frac{\sum_{b=1}^2 c_{ib} h_{ib}^M}{\max(\sum_{b=1}^2 c_{ib}, \varepsilon)}, \quad \bar{c}_i = \tfrac{1}{2}(c_{i1}+c_{i2}), \\ a_i^M &= \mathrm{sigmoid}\{W_g(h_i^D \parallel h_i^M \parallel c_{i1} \parallel c_{i2}) + b_g\}, \\ h_i &= h_i^D + \bar{c}_i\, a_i^M \odot W_M h_i^M. \end{aligned} \tag{S3}$$

Thus missing monthly coverage continuously reduces the contextual correction. Missingness is represented by explicit channels and coverage statistics; zero-imputed observations are not automatically treated as attention-padding positions.

B. A factorized cross-stock relation operator

Set Transformer uses learned inducing points to aggregate a set [11], and Slot Attention uses competition across slots followed by a weighted mean over inputs [12]. In our operator, stocks write to shared summaries by this latter rule and read from them with an independently learned projection. Let $C = 64$ and let $e_c \in \mathbb{R}^{32}$ be learned summary keys. Stock-to-summary writing and summary-to-stock reading use separate projections:

$$\begin{aligned} A_{ic} &= \mathrm{softmax}_c\{\mathrm{unit}(W_K h_i)^\top \mathrm{unit}(e_c)\}, \\ B_{ic} &= \mathrm{softmax}_c\{\mathrm{unit}(W_R h_i)^\top \mathrm{unit}(e_c)\}, \\ z_c &= \frac{\sum_i v_i A_{ic} W_V h_i}{\max(\sum_i v_i A_{ic}, \varepsilon)}, \\ r_i^0 &= v_i \left(h_i + W_O \sum_c B_{ic} z_c \right). \end{aligned} \tag{S4}$$

Writing aggregates information from the full available stock universe; reading adapts the mixture of aggregate states for each stock. This operation materializes $N \times C$ assignments instead of an $N \times N$ relation matrix.

Proposition 1 (Effective stock relations): When all summary masses are at least ε, the summary readout is a factorized stock-to-stock aggregation $\sum_c B_{ic} z_c = \sum_\ell \Omega_{i\ell} W_V h_\ell$, where

$$\Omega_{i\ell} = \sum_c B_{ic} \frac{v_\ell A_{\ell c}}{\sum_a v_a A_{ac}}. \tag{S5}$$

Each valid row of Ω is nonnegative and sums to one, and $\mathrm{rank}(\Omega) \le C$.

Proof: Substitution of z_c gives the identity. Summing over ℓ cancels each summary denominator, leaving $\sum_c B_{ic} = 1$. Finally, Ω is the product of an $N \times C$ matrix and a $C \times N$ matrix. The standard product-rank bound [13] therefore gives $\mathrm{rank}(\Omega) \le C$. ∎

S3. Allocation and Holdings-Conditioned Diagnosis

A. Initial signed allocation

AlphaStock and AlphaPortfolio use stock scores to form long and short portfolios [14], [6]. Here those weights initialize the candidate whose holdings and available capital guide refinement. A scalar head gives $s_i = f_{\mathrm{score}}(r_i^0)$. Let $\mathcal{L}$ and $\mathcal{S}$ denote configured long and short admissible supports. Eligibility is evaluated from decision-time information. The daily side owner $o_i \in \{+1, -1\}$ is fixed during refinement; separate admission masks exclude inadmissible stocks. Stable permanent-ID tie breaking makes rank-based admission deterministic. Given one eligible set per side, the candidate allocation is

$$\begin{aligned} p &= \mathrm{softmax}_{\mathcal{L}}(s), \qquad n = \mathrm{softmax}_{\mathcal{S}}(-s), \\ w^0 &= \beta^+ p - \beta^- n. \end{aligned} \tag{S6}$$

The budget head uses the mean fused state and its long/short weighted pools, $z^{\mathrm{bud}} = \bar{h} \parallel \sum_i p_i h_i \parallel \sum_i n_i h_i$:

$$\begin{aligned} \beta^+ = \beta &= 0.02 + 0.96\,\mathrm{sigmoid}\{f_{3d\to 32\to 1}^{\tanh}(z^{\mathrm{bud}})\}, \\ \beta^- &= 1 - \beta. \end{aligned} \tag{S7}$$

With nonempty disjoint supports, gross exposure is one and signed exposure is $2\beta - 1$. An empty support leaves that side's budget in cash instead of increasing the other side's exposure.

B. Holdings-aware relational states

Write $w_i^+ = \max(w_i, 0)$, $w_i^- = \max(-w_i, 0)$, and $B^\pm = \sum_i w_i^\pm$. At round k, define

$$\begin{aligned} c^\pm &= \frac{\sum_i w_i^\pm r_i^k}{\max(B^\pm, \varepsilon)}, \quad \bar{r} = \frac{\sum_i v_i r_i^k}{\max(\sum_i v_i, 1)}, \\ z^{\mathrm{ctx}} &= c^+ \parallel c^- \parallel B^+ \parallel B^- \parallel \sum_i w_i^2 \\ &\quad \parallel \frac{\sum_i \mathbf{1}\{|w_i| > 10^{-10}\}}{\max(\sum_i v_i, 1)}, \qquad c = W_c z^{\mathrm{ctx}} + b_c. \end{aligned} \tag{S8}$$

Let $H_i = \mathbf{1}\{|w_i| > 10^{-10}\}$ and $B_i = B^+$ or B^- according to o_i. Define $a_i^{\mathrm{ctr}} = \mathbf{1}\{i \text{ is admissible}, B_i - |w_i| > \varepsilon\}$ and probe size $\nu_i = \frac{1}{4} a_i^{\mathrm{ctr}} \{H_i |w_i| + (1 - H_i) B_i\}$. The 12-coordinate descriptor ρ_i contains

$$\begin{aligned} \rho_i = \big(&\log(1 + N w_i^+), \log(1 + N w_i^-), \\ &\mathbf{1}\{w_i > 0\}, \mathbf{1}\{w_i < 0\}, o_i, |w_i| / \max(B_i, \varepsilon), \\ &\tanh \tilde{s}_i, \tanh\{N(w_i^k - w_i^{k-1})\}, B_i, \\ &w_i^2 / \max(\sum_\ell w_\ell^2, \varepsilon), a_i^{\mathrm{ctr}}, \nu_i \big). \end{aligned} \tag{S9}$$

Here $\tilde{s}_i$ is the cross-sectional standardized score, using population variance with floor 10^{-12}, and the first-round weight

change is zero. The conditioned state and contribution prediction are

$$\begin{aligned} z_i^{\mathrm{cond}} &= r_i^k \parallel (r_i^k - c^+) \parallel (r_i^k - c^-) \\ &\quad \parallel (r_i^k - \bar{r}) \parallel c \parallel \rho_i \parallel x_i \parallel b_i, \\ \tilde{r}_i &= r_i^k + f_{\mathrm{cond}}(z_i^{\mathrm{cond}}), \\ \hat{y}_i &= a_i^{\mathrm{ctr}} \kappa \tanh(w_y^\top \tilde{r}_i + b_y), \\ \hat{r}_i &= v_i\{\tilde{r}_i + W_y(\hat{y}_i/\kappa \parallel \nu_i \parallel a_i^{\mathrm{ctr}})\}, \qquad \kappa = 0.02. \end{aligned} \tag{S10}$$

The input width of f_{cond} is $5d+12+2S = 1434$. Its output is a residual in the original relation-state space. The contribution label is derived in Section S7.

C. Diagnostic queries and capacity-limited sources

Pooling with learned queries follows attention-based set pooling [11]. We condition each query on the current portfolio and give it a side, funding positions and a per-request capital limit. Each learned diagnostic embedding e_j^D is assigned a side: 16 requests are long-side and 16 short-side. Let $t_j = e_j^D + \hat{c}$, where $\hat{c}$ is the portfolio context recomputed from $\hat{r}$. With $\ell_{ji} = t_j^\top K_D \hat{r}_i/\sqrt{d}$,

$$\begin{aligned} \alpha_{ji}^{\mathrm{pool}} &= \mathrm{softmax}_{\mathcal{V}}(\ell_j)_i, \\ \alpha_{ji}^{\mathrm{src}} &= \mathrm{softmax}_{\{i:o_i=\sigma_j, H_i=1\}}\{\ell_{ji} + \log|w_i|\}_i. \end{aligned} \tag{S11}$$

The source weights balance learned relevance and deployed capital. The Gibbs variational identity [4], applied with the normalized source holdings as the reference measure, shows that they maximize $\sum_i \alpha_i \ell_{ji} - \mathrm{KL}(\alpha \| p^{\mathrm{cap}})$ over the source simplex, where $p_i^{\mathrm{cap}} = |w_i| / \sum_{\ell \in \mathrm{src}} |w_\ell|$. The Lagrangian derivative $\ell_{ji} - \log(\alpha_i/p_i^{\mathrm{cap}}) - 1 - \lambda = 0$ gives $\alpha_i \propto p_i^{\mathrm{cap}} \exp(\ell_{ji})$.

Define pooled states $h_j^{\mathrm{pool}} = \sum_i \alpha_{ji}^{\mathrm{pool}} \hat{r}_i$ and $h_j^{\mathrm{src}} = \sum_i \alpha_{ji}^{\mathrm{src}} \hat{r}_i$. For attribute f, its known mass is $k_{jf} = \sum_i \alpha_{ji}^{\mathrm{src}} b_{if}$ and its observed mean is $\bar{x}_{jf} = \sum_i \alpha_{ji}^{\mathrm{src}} b_{if} x_{if} / \max(k_{jf}, \varepsilon)$. Then

$$\begin{aligned} \psi_j &= f_{\mathrm{diag}}(t_j \parallel h_j^{\mathrm{pool}} \parallel h_j^{\mathrm{src}} \parallel \bar{x}_j \parallel k_j), \\ q_j &= \mathrm{unit}(W_q \psi_j). \end{aligned} \tag{S12}$$

Linear heads additionally predict an action-intent logit and observed source attributes. The source distribution u_j retains and renormalizes the four largest source weights. Its capital limit is

$$m_j = \frac{1}{4} \min_{i:u_{ji}>0} \frac{|w_i|}{u_{ji}}. \tag{S13}$$

For every source with $u_{ji} > 0$, (S13) gives $m_j u_{ji} \le |w_i|/4$. Thus one full plan consumes at most a quarter of each included source position before the solver accounts for overlapping requests. A request with no source produces no transfer plan.

S4. Budgeted Access and Transfer-Plan Construction

A. Hierarchical access

Stock i enters summary group $\arg\max_c A_{ic}$. Within a group we use the axis-aligned median partition of a k-d tree [15]. Retrieval keys $\mathrm{unit}(W_{\mathrm{ret}} r_i)$ are recursively split at the median along their largest-variance coordinate until each leaf contains at most 64 stocks. Tree topology is fixed for the decision; updates to touched stocks refresh their ancestor summaries. A node stores its mean relation state and a normalized aggregate key. For request j, an eligible target is valid, admitted, assigned owner σ_j, and absent from its own source support.

An access network combines the request, node, and current search state:

$$\phi_{jv} = \psi_j \parallel h_v \parallel (q_j \odot q_v) \parallel \xi_j \parallel f^{\mathrm{glob}} \parallel \chi_{jv}. \tag{S14}$$

The dimensions are $256 + 256 + 32 + 16 + 8 + 9 = 577$. The nine coordinates of χ are eligible fraction, node size divided by N, already-read fraction, depth divided by 16, newly required coarse and fine work divided by their budgets, m_j, and remaining coarse and fine budget fractions. We combine the ensemble mean with a disagreement bonus, following the exploration rule of Chen et al. described by Lee et al. [16, Eq. (7)]. Cost-aware acquisition also ranks an evaluation by its expected benefit per unit cost [17]. Here the benefit is a change in feasible portfolio utility, and the cost counts new reads and comparisons. Three heads share a 577-to-256 GELU trunk and predict values $\kappa \tanh(\cdot)$. If their mean and population standard deviation are $\bar{v}_{jv}, s_{jv}$, the priority is

$$\mathrm{priority}(j, v) = \frac{\bar{v}_{jv} + 0.25 s_{jv}}{n_c + 4 n_f + 32}, \tag{S15}$$

where n_c, n_f estimate newly required coarse and fine work. Previously read edges are cached. A fixed fraction of access is reserved for reproducible exploration keyed by decision date and update index. Budgets cover all requests and rounds of the decision, not each request independently: coarse reads 16,384, node visits 32,768, and fine edges 1,024.

B. Two-wave search and solver feedback

This is the main paper's *access feedback*: it changes which evidence is inspected, not the candidate or stock states used by that round. Each round receives a share of the remaining budget. The first wave uses approximately half of that round's allowance, leaving resources for feedback-guided expansion. The first wave constructs plans and invokes the solver described in Section S5. The second wave uses its feedback to inspect additional targets and append plan versions. The current allocation, conditioned states, and diagnostic requests remain fixed throughout both waves; writeback occurs only after the second solve. Consequently, the second wave can compare additional directions against a retained feasible incumbent within the same local problem.

The feedback vector ξ_j has 16 coordinates: inspected and eligible target counts divided by 128; source capacity; remaining fine-budget fraction; a plan-existence indicator; accepted coefficient sum for this request; source-weighted utilization and remaining capacity; scaled conflict, gain, and revision-penalty-gradient summaries; primal and dual residuals; a convergence flag; objective improvement divided by κ; and

revision penalty divided by κ. For coefficient vector a, utilization is $(C_{\mathrm{src}}a)_i/\max(|w_i|,\varepsilon)$, conflict is Qa, and a revision-penalty subgradient is $\lambda_{\mathrm{rep}}D^\top \operatorname{sign}(Da)$, taking $\operatorname{sign}(0) = 0$. The scale of the request-level conflict/gain/penalty terms is $\max(m_j\kappa, 10^{-6})$ followed by $\tanh$. The global eight-dimensional feedback is $f^{\mathrm{glob}} = F^\top a$. Feedback into the access policy is stop-gradient.

An audit label measures the feasible local utility change after adding one inspected plan:

$$y^{\mathrm{access}} = \widehat{U}(\mathcal{P} \cup \{p_{\mathrm{new}}\}) - \widehat{U}(\mathcal{P}). \tag{S16}$$

Both problems must pass feasibility and convergence checks. Audit solves at 256 and 512 iterations must agree on the increment within 10^{-6}. At most two audits are used per round. This supervises the utility of an inspected action, not an unobserved node's eventual return. The three access heads use independent 0.8 inclusion masks following the bootstrapped-head training scheme of Osband et al. [18], with at least one head assigned each label.

C. Fine target distributions

Let $\mathcal{E}_p$ be the visited eligible support for a plan attached to request $j = j(p)$, and define $h_j^{\mathrm{src}} = \sum_i u_{ji}\hat{r}_i$. The four-coordinate holding vector is $b_i^H = (\log(1 + Nw_i^+), \log(1 + Nw_i^-), \mathbf{1}\{w_i > 0\}, \mathbf{1}\{w_i < 0\})$. Let $z_{ji}^{\mathrm{marg}} = \psi_j \parallel (\hat{r}_i - h_j^{\mathrm{src}}) \parallel b_i^H \parallel x_i \parallel b_i$. The fine comparison uses scaled dot-product attention [7]. RECAST adds source context, holding-dependent projections and the coarse retrieval score. Fine keys, values, and query are

$$\begin{aligned}
k_i^f &= W_K^f \hat{r}_i + W_K^H b_i^H, \quad v_i^f = W_V^f \hat{r}_i + W_V^H b_i^H, \\
q_j^f &= W_Q^f(\psi_j \parallel h_j^{\mathrm{src}}), \\
\ell_{pi}^f &= (q_j^f)^\top k_i^f/\sqrt{32} + \ell_{ji}^{\mathrm{coarse}} + f_{\mathrm{marg}}(z_{ji}^{\mathrm{marg}}), \\
\pi_p &= \operatorname{softmax}_{\mathcal{E}_p}(\ell_p^f), \qquad h_p^{\mathrm{dst}} = \sum_i \pi_{pi} v_i^f.
\end{aligned} \tag{S17}$$

The coarse score combines retrieval-key similarity with the predicted access value divided by κ. The marginal head has input width $2d + 4 + 2S = 658$. Empty target supports yield no plan. An admissible plan is

$$\begin{aligned}
d_p &= \sigma_j m_j(\pi_p - u_j), \\
g_p &= m_j\kappa \tanh\{f_{\mathrm{gain}}(\psi_j \parallel h_p^{\mathrm{dst}} \parallel h_j^{\mathrm{src}})\} \operatorname{sigmoid}(\mathrm{intent}_j), \\
f_p &= m_j\{W_F(h_p^{\mathrm{dst}} \parallel h_j^{\mathrm{src}}) + b_F\}.
\end{aligned} \tag{S18}$$

Each direction moves capital within one side; $\mathbf{1}^\top d_p = 0$. Two waves can produce up to two versions per request, so the plan library can contain $P \le 2J = 64$ columns. The limit of 32 active queries does not imply a final library of at most 32 columns.

S5. Coupled Acceptance and Feasible Optimization

A. One joint capital-allocation problem

Convex portfolio models commonly combine a linear gain, a quadratic penalty and trading costs [19]. Our variables are acceptance fractions for proposed transfers. Their quadratic terms measure overlap between these proposals, and their linear constraints enforce shared source capacity. Transfer coefficients $a \in \mathbb{R}^P$ are selected jointly rather than by independent plan thresholds:

$$\begin{aligned}
U(a) &= g^\top a - \tfrac{1}{2}a^\top Q a - \lambda_{\mathrm{rep}}\|Da\|_1, \\
Q &= FF^\top + \lambda_D D^\top D + \varepsilon_T T_s^\top T_s, \\
\mathcal{A} &= \{a : 0 \le a \le 1,\ C_{\mathrm{src}}a \le |w|, \\
&\qquad\qquad T_s a \le \mathbf{1},\ Ga \le q\}, \\
(C_{\mathrm{src}})_{ip} &= (-o_i D_{ip})_+, \qquad \lambda_D = 0.05, \quad \varepsilon_T = 10^{-6}.
\end{aligned} \tag{S19}$$

The matrix C_{src} counts withdrawals from existing side-owned positions. The request constraint prevents multiple versions of one request from jointly exceeding its source budget. Additional rows $Ga \le q$ express linear portfolio constraints relative to the current feasible allocation, so $q \ge 0$ and $a = 0$ is feasible. All rows with zero capacity are handled explicitly.

The three quadratic terms penalize learned factor overlap, overlap between signed stock transfers, and multiple versions of one request. The revision regularizer uses the net within-round change Da. Opposing internal transfers can offset before penalization, since $\|\sum_p a_p d_p\|_1 \le \sum_p a_p \|d_p\|_1$. This term moderates candidate changes inside the learned local problem. Only the final target is submitted to the execution ledger, which computes fees from the resulting actual trades; internal revisions are not booked as executed transactions.

For our transfer directions, the source-capacity constraints give a per-stock bound on withdrawals. Conservation of side capital then follows from the zero column sum in (S18).

Proposition 2 (Feasible side-preserving repair): Suppose w respects the fixed owner mask, each plan moves capital only between stocks with owner $\sigma_{j(p)}$, and both its source and target distributions sum to one. For $a \in \mathcal{A}$, $w' = w + Da$ has no sign reversal and preserves the total capital on each side.

Proof: Let $B_D = (\operatorname{diag}(o)D)_+$ and note $\operatorname{diag}(o)D = B_D - C_{\mathrm{src}}$. Since $o_i w_i = |w_i|$ on admissible stocks,

$$\operatorname{diag}(o)w' = |w| + (B_D - C_{\mathrm{src}})a \ge B_D a \ge 0. \tag{S20}$$

Thus no position crosses to the other side. Each column has support on one side and sums to zero, so that side's signed total is unchanged. Together with sign preservation, this also preserves its absolute capital. Inadmissible stocks have zero rows in D. ∎

The next result applies standard convexity rules for positive semidefinite quadratic forms and norms [20] to our plan matrix. The identity block in our splitting makes the linear update of the alternating direction method of multipliers (ADMM) [21] uniquely solvable.

Proposition 3 (Convexity and a unique ADMM linear solve): The negative utility in (S19) is convex on the polyhedral feasible set. Its optimizer need not be unique. The normal matrix used by the ADMM coefficient update below is positive definite.

Proof: For any $z \in \mathbb{R}^P$,

$$z^\top Q z = \|F^\top z\|_2^2 + \lambda_D \|Dz\|_2^2 + \varepsilon_T \|T_s z\|_2^2 \geq 0. \quad \text{(S21)}$$

The remaining terms are linear or convex norms. The stacked constraint operator used below contains the identity, so $A_s^\top A_s \succeq I$. Hence $Q/\gamma + \rho A_s^\top A_s \succeq \rho I \succ 0$ for $\rho > 0$. ∎

B. Internal revisions and execution costs

Write $w = w^k$ for the current virtual candidate and $\Delta w = Da$ for its proposed revision. The regularizer $P_{\mathrm{rep}}(a) = \lambda_{\mathrm{rep}} \|\Delta w\|_1$ is centered on w^k and penalizes the magnitude of the candidate revision.

A linear-cost comparison makes the distinction precise. Fix actual pre-trade holdings w^{pre} and express all weights relative to the same pre-trade equity. Under the proportional trading-cost model [19], with complete fills and a uniform rate c_f, define the reference execution cost and its change by

$$\begin{aligned} C_{\mathrm{lin}}(v; w^{\mathrm{pre}}) &= c_f \|v - w^{\mathrm{pre}}\|_1, \\ \Delta C_{\mathrm{lin}}(a) &= C_{\mathrm{lin}}(w^k + Da; w^{\mathrm{pre}}) \\ &\quad - C_{\mathrm{lin}}(w^k; w^{\mathrm{pre}}). \end{aligned} \quad \text{(S22)}$$

This signed difference captures both cost increases and savings.

Proposition 4 (Linear-cost variation bound): Under the reference assumptions in (S22),

$$|\Delta C_{\mathrm{lin}}(a)| \leq c_f \|Da\|_1. \quad \text{(S23)}$$

For the implemented $\lambda_{\mathrm{rep}} = c_f$, $P_{\mathrm{rep}}(a)$ bounds the absolute variation in this reference execution cost.

Proof: Set $e^k = w^k - w^{\mathrm{pre}}$. The bound is the reverse triangle inequality for the ℓ_1 norm [20, App. A], applied to the trade vectors before and after revision. To see this, write

$$\begin{aligned} \|e^k + \Delta w\|_1 &\leq \|e^k\|_1 + \|\Delta w\|_1, \\ \|e^k\|_1 &\leq \|e^k + \Delta w\|_1 + \|\Delta w\|_1. \end{aligned}$$

Subtracting $\|e^k\|_1$ in the first inequality and rearranging the second bounds the norm difference between $-\|\Delta w\|_1$ and $\|\Delta w\|_1$. Multiplication by $c_f \geq 0$ proves (S23). ∎

For example, keep all other positions unchanged and let actual long weights in stocks A and B be $(0.30, 0.30)$. A candidate $(0.40, 0.20)$ incurs a reference fee of 1 bp when $c_f = 0.0005$. Revising it back to $(0.30, 0.30)$ reduces that fee to zero, so $\Delta C_{\mathrm{lin}} = -1$ bp while $P_{\mathrm{rep}} = +1$ bp. Thus, P_{rep} provides a conservative revision threshold under the stated linear reference. The execution ledger applies the configured trading and settlement rules and supplies actual net returns to the path objective. P_{rep} serves solely as a local optimization penalty, without an additional charge to account cash or returns.

C. Scaled ADMM and feasibility polishing

Normalize the upper-bound constraints into $La \leq b$; for example, positive-capacity source rows are $(C_{\mathrm{src}})_{i:}/|w_i|$ with bound one. Bounds remain nonnegative. A detached scalar

$$\begin{aligned} \gamma = \max\{ &\max_p |g_p|,\ \max_{pq} |Q_{pq}|, \\ &\lambda_{\mathrm{rep}} \max_p \|d_p\|_1,\ 10^{-5}\} \end{aligned} \quad \text{(S24)}$$

normalizes the objective without changing its minimizers. Set $\tilde{Q} = Q/\gamma$, $\tilde{g} = g/\gamma$, $\tilde{\lambda} = \lambda_{\mathrm{rep}}/\gamma$, and $A_s = [I^\top, L^\top, D^\top]^\top$. The split problem minimizes $\frac{1}{2} a^\top \tilde{Q} a - \tilde{g}^\top a + h(z)$ subject to $A_s a = z$, where $h(z) = I_{[0,1]^P}(z_B) + I_{\{z_L \leq b\}}(z_L) + \tilde{\lambda}\|z_D\|_1$. With scaled dual variable u, ADMM [21] gives the following updates. The three z blocks use box projection, componentwise upper-bound projection and the ℓ_1 soft-threshold operator, respectively [22].

$$\begin{aligned} (\tilde{Q} + \rho A_s^\top A_s) a^{\ell+1} &= \tilde{g} + \rho A_s^\top (z^\ell - u^\ell), \\ v^{\ell+1} &= A_s a^{\ell+1} + u^\ell, \\ z_B^{\ell+1} &= \mathrm{clip}(v_B^{\ell+1}, 0, 1), \\ z_L^{\ell+1} &= \min(v_L^{\ell+1}, b), \\ z_D^{\ell+1} &= \mathrm{sign}(v_D^{\ell+1})(|v_D^{\ell+1}| - \tilde{\lambda}/\rho)_+, \\ u^{\ell+1} &= u^\ell + A_s a^{\ell+1} - z^{\ell+1}. \end{aligned} \quad \text{(S25)}$$

The reference forward solve uses 48 iterations and $\rho = 0.5$. Primal and dual residuals are $r_{\mathrm{pri}} = A_s a - z$ and $r_{\mathrm{dual}} = \rho A_s^\top (z^\ell - z^{\ell-1})$. Convergence is checked in the infinity norm at tolerance 10^{-4}; finite iteration counts are not treated as exact minimization.

After the finite ADMM solve, our feasibility step restricts the candidate to a ray. First clip the coefficient vector $\bar{a} = \mathrm{clip}(a, 0, 1)$, then shrink toward the feasible origin:

$$\eta = \min\left\{1, \min_{j:(L\bar{a})_j > b_j} \frac{b_j}{(L\bar{a})_j}\right\}, \qquad a^{\mathrm{pol}} = \eta \bar{a}. \quad \text{(S26)}$$

Each violated row requires $\eta(L\bar{a})_j \leq b_j$ and has a positive left-hand side before scaling. The minimum gives the largest feasible fraction $\eta \in [0, 1]$ of the clipped candidate. Because $b_j \geq 0$, every other row also remains satisfied after shrinking. The implementation checks finite values and constraint residuals, and retains the highest-utility feasible candidate among the polished solution, zero repair, and an available incumbent.

Within a round, the old library embeds in the expanded one by assigning zero coefficients to new plans. The guarantee below follows from this inclusion and the solver's explicit comparison with the incumbent.

Proposition 5 (Same-round incumbent preservation): Append plans to a fixed-round library while retaining all old directions, gains, and the old principal block of Q. If the incumbent $a^\star$ was feasible, then its zero extension $(a^\star, 0)$ is feasible with unchanged utility. Selecting the best verified feasible candidate therefore preserves incumbent utility, up to numerical tolerance.

Proof: The added zero coefficients contribute neither withdrawals nor request mass, stock transfers, or linear-constraint changes. The old principal block gives the same

quadratic form, and all other objective terms are unchanged. The candidate comparison explicitly includes this extension. ∎

The result applies to the fixed local utility within one round; writeback defines the local problem for the next round.

S6. Relation Writeback and Within-Decision Stopping

This is the main paper's *state feedback*: after both retrieval waves, the accepted allocation and relational messages change the starting state of the next diagnosis. It is separate from access feedback in S4.B.

A. From accepted plans to stock states

We use the message-and-aggregate form of neural message passing [23] on the stock–plan incidences. Each message encodes the role of the stock and the capital assigned by the joint solver. After acceptance, $w^{k+1} = w^k + D\hat{a}$. For stock i participating in plan p, let $\epsilon_{ip} = +1$ for a target and -1 for a source, and $\omega_{ip} = \pi_{pi}$ or $u_{j(p)i}$, respectively. Let $\mu_{ip} = m_{j(p)}\omega_{ip}$. The counterpart state is $h^{\mathrm{src}}_{j(p)}$ for a target and h^{dst}_{p} for a source. The plan-to-stock message is

$$\begin{aligned} z^{\mathrm{msg}}_{ip} &= r^k_i \,\|\, h^{\mathrm{other}}_{ip} \,\|\, \psi_{j(p)} \,\|\, \epsilon_{ip}\sigma_{j(p)} \\ &\quad \|\, N\mu_{ip} \,\|\, N\mu_{ip}\hat{a}_p \,\|\, \frac{g_p}{\max(m_{j(p)}\kappa, \varepsilon)}, \\ M_{ip} &= f_{\mathrm{msg}}(z^{\mathrm{msg}}_{ip}). \end{aligned} \tag{S27}$$

The message distinguishes proposed capital, accepted capital, and role. Rejected proposals can still contribute diagnostic context through their zero accepted-capital coordinate. Pool messages as $\bar{M}_i = \sum_p \omega_{ip} M_{ip} / \max(\omega_i, \varepsilon)$, where $\omega_i = \sum_p \omega_{ip}$. Define

$$\begin{aligned} \zeta_i &= (r^k_i - r^0_i) \,\|\, (\hat{r}_i - r^k_i) \,\|\, \bar{M}_i \\ &\quad \|\, \tanh\{N(w^{k+1}_i - w^k_i)\} \,\|\, \log(1+\omega_i), \\ g^W_i &= \mathrm{sigmoid}(W_{\mathrm{gate}}\zeta_i + b_{\mathrm{gate}}), \\ v^W_i &= \tanh\{f_{\mathrm{write}}(\zeta_i)\}, \\ r^{k+1}_i &= r^k_i + 0.25\, v_i \mathbf{1}\{\omega_i > 0\}\, g^W_i \odot v^W_i. \end{aligned} \tag{S28}$$

The input width is $3d + 2 = 770$. Since each sigmoid gate is at most one and $|\tanh x| \le 1$, (S28) bounds the magnitude of every coordinate change by 0.25, and untouched stock states are unchanged. The next diagnostic round therefore operates on states informed by both the candidate book and the transfers jointly accepted for that book.

B. Stopping and temporal scope

Early stopping requires two consecutive rounds satisfying all of: feasible and converged solver diagnostics; $\|w^{k+1} - w^k\|_1 \le 10^{-4}$; maximum relative touched-state change at most 10^{-3}; maximum relative diagnostic-state change at most 10^{-3}; and remaining opportunity at most 10^{-6}. The opportunity statistic is the maximum of the positive plan-gain sum and the sum, over requests, of their largest positive frontier value. It is treated as unknown if node budget is exhausted with a nonempty frontier, rather than equated to zero. Independent termination conditions are the four-round cap, exhausted access budget, or absence of admissible proposals.

Refinement is internal to one decision, not a sequence of executed intraday trades. The final allocation is sent to the execution layer. The next decision recomputes neural states from its input windows, while actual positions and cash carry through the ledger.

S7. Training Labels and Auxiliary Objectives

A. Counterfactual marginal contribution

We construct the contribution label by moving a small amount of capital within the same side of the candidate and measuring the resulting change in log wealth. Training outcomes are supplied to loss construction only. Let z_i be the stock return over the label interval, available only after the decision. For a valid label, define

$$\begin{aligned} R_0 &= \sum_i w_i z_i, \\ z^{\mathrm{ref}}_i &= \frac{\sum_{\ell: o_\ell = o_i} |w_\ell| z_\ell - |w_i| z_i}{B_i - |w_i|}, \\ \Delta R_i &= o_i \nu_i (z_i - z^{\mathrm{ref}}_i). \end{aligned} \tag{S29}$$

This is a same-side capital transfer: the probe removes or adds ν_i in stock i and allocates the counterpart proportionally to the remaining side book. The two legs produce incremental fee $2c_f\nu_i$ under the local linear-cost label convention. The counterfactual local return and per-unit log contribution are

$$\begin{aligned} R^{\mathrm{cf}}_i &= \begin{cases} R_0 - \Delta R_i - 2c_f\nu_i, & H_i = 1, \\ R_0 + \Delta R_i - 2c_f\nu_i, & H_i = 0, \end{cases} \\ y_i &= \begin{cases} [\log(1+R_0) - \log(1+R^{\mathrm{cf}}_i)]/\nu_i, & H_i = 1, \\ [\log(1+R^{\mathrm{cf}}_i) - \log(1+R_0)]/\nu_i, & H_i = 0. \end{cases} \end{aligned} \tag{S30}$$

Only positive probes with valid reference mass, valid return labels, and positive log arguments enter this loss. These are local counterfactual labels; the complete strategy objective uses the execution ledger.

The construction follows from log-wealth differences. For example, when i is held, $1 + R^{\mathrm{cf}}_i = 1 + R_0 - \nu_i\{o_i(z_i - z^{\mathrm{ref}}_i) + 2c_f\}$. A first-order Taylor expansion with remainder [24], $\log(x - \delta) = \log x - \delta/x + O(\delta^2)$, yields

$$y_i = \begin{cases} \dfrac{o_i(z_i - z^{\mathrm{ref}}_i) + 2c_f}{1 + R_0} + O(\nu_i), & H_i = 1, \\ \dfrac{o_i(z_i - z^{\mathrm{ref}}_i) - 2c_f}{1 + R_0} + O(\nu_i), & H_i = 0. \end{cases} \tag{S31}$$

The difference in fee sign is economically meaningful: retaining a held stock avoids the replacement fee, whereas adding a new one must earn that fee. Labels and validity masks are detached.

B. Ranking, gain, observation, and access supervision

We use Huber's loss [25] with unit threshold. It is $H(x) = x^2/2$ for $|x| \le 1$ and $H(x) = |x| - 1/2$ otherwise. All auxiliary averages below are over valid examples. Contribution supervision is $L_{\mathrm{ctr}} = \mathrm{mean}_i\, H((\hat{y}_i - \mathrm{sg}(y_i))/\kappa)$.

The ranking term uses the top-one listwise cross-entropy of Cao et al. [26]. Its comparison set here consists of stocks inspected for a particular source request. For an inspected candidate set with at least two valid future returns, the side-aware target distribution is $q_i^{\mathrm{lab}} = \mathrm{softmax}(\sigma_j z_i/0.01)$. Both fine and coarse candidate probabilities receive

$$L_{\mathrm{rank}} = -\,\mathrm{mean}_{\mathrm{sets}} \sum_i q_i^{\mathrm{lab}} \{\log P_i^{\mathrm{fine}} + \log P_i^{\mathrm{coarse}}\}. \quad \text{(S32)}$$

This ranks targets for the same source request; it does not compare unmatched capital budgets. The realized gross transfer label is $g_p^{\mathrm{lab}} = \mathrm{sg}(d_p^\top z)$. Set $\bar{m}_p = \max\{\mathrm{sg}(m_{j(p)}), 10^{-8}\}$. The gain regression uses $H((g_p - g_p^{\mathrm{lab}})/(\bar{m}_p\kappa))$. The intent classifier targets $\mathbf{1}\{g_p^{\mathrm{lab}} - 2c_f\bar{m}_p > 0\}$. The factor two counts the source and target legs of the disjoint transfer. Their equally weighted average defines L_{value}.

The observation head reconstructs the source's observed attributes using known-mass weighting,

$$L_{\mathrm{obs}} = \frac{\sum_{j,f} \mathrm{sg}(k_{jf})(\hat{x}_{jf} - \mathrm{sg}(\bar{x}_{jf}))^2}{\max(\sum_{j,f} \mathrm{sg}(k_{jf}), 1)}. \quad \text{(S33)}$$

An attention-row overlap penalty was used by Lin et al. [27] to encourage heads to attend to different inputs. The diversity term averages squared cosine similarities between different normalized diagnostic pooling distributions α_j^{pool}, encouraging distinct diagnostic reads. Access supervision applies $H((\hat{v} - \mathrm{sg}(y^{\mathrm{access}}))/\kappa)$ to audited head-label pairs. The complete per-decision auxiliary loss is

$$\begin{aligned} L_{\mathrm{aux}} = {} & 0.10L_{\mathrm{rank}} + 0.02L_{\mathrm{obs}} + 0.05L_{\mathrm{value}} \\ & + 0.001L_{\mathrm{div}} + 0.05L_{\mathrm{ctr}} + 0.05L_{\mathrm{access}}. \end{aligned} \quad \text{(S34)}$$

Feature reconstruction, target ordering, marginal contribution, plan gain, and search value supervise different interfaces of the same forward computation. Future outcomes never become forward retrieval features or decision-time eligibility inputs.

S8. Execution-Aware Path Training and Replay

A. Execution, cost, and economic objective

Distinguish the neural target w_t, pre-trade actual weights w_t^{pre} after market drift, and executed weights w_t^{exe}. They use a common pre-trade net-asset-value denominator. The execution layer applies tradability, availability, and event processing before updating positions and cash. Let δ_t denote the signed disposal amounts, in the same pre-trade-equity units, recorded by the terminal-settlement and valuation policy. The rebalancing fee and the total fee including settlement are

$$\begin{aligned} C_t^{\mathrm{rebal}} &= c_f \|w_t^{\mathrm{exe}} - w_t^{\mathrm{pre}}\|_1, \\ C_t^{\mathrm{total}} &= C_t^{\mathrm{rebal}} + c_f \|\delta_t\|_1. \end{aligned} \quad \text{(S35)}$$

These charges are computed from actual trades and settlements. The ledger values actual holdings and cash under the configured fee, valuation, and corporate-action rules to obtain net returns $R_t = V_t/V_{t-1} - 1$. When an order remains unfilled, the existing holding continues to be valued in the account.

Moody and Saffell train trading policies with risk-adjusted performance and transaction costs [28], and AlphaPortfolio directly optimizes portfolio objectives [6]. We measure the executed return path using the Sharpe ratio [29] and log-wealth growth [30]. Let L_{econ} denote the economic path loss in Sec. IV-E of the main paper. For a path of T decisions, L_{ret} denotes the component derived from realized net returns.

$$\begin{aligned} \bar{R} &= T^{-1} \sum_t R_t, \\ s_R &= \max\left\{ \sqrt{T^{-1} \sum_t (R_t - \bar{R})^2}, 10^{-6} \right\}, \\ S_{\mathrm{tr}} &= \sqrt{252}\, \bar{R}/s_R, \qquad G_{\log} = \frac{252}{T} \sum_t \log(1 + R_t), \\ L_{\mathrm{ret}} &= -S_{\mathrm{tr}} - 0.25 G_{\log}, \\ L &= L_{\mathrm{econ}} + T^{-1} \sum_t L_{\mathrm{aux},t}. \end{aligned} \quad \text{(S36)}$$

Here S_{tr} is the main paper's S_{net}. The log-growth weight is $\lambda_G = 0.25$, and the standard-deviation floor is $\epsilon_R = 10^{-6}$. The differentiable training objective uses the population standard deviation, and reported Sharpe ratios use the sample standard deviation. Both use a zero risk-free rate. $G_{\log}$ is annualized log growth; annualized compound return is $\exp(G_{\log}) - 1$.

For the unfloored standard-deviation branch, $\partial s_R/\partial R_t = (R_t - \bar{R})/(T s_R)$. Consequently,

$$\begin{aligned} \frac{\partial S_{\mathrm{tr}}}{\partial R_t} &= \sqrt{252} \left\{ \frac{1}{T s_R} - \frac{\bar{R}(R_t - \bar{R})}{T s_R^3} \right\}, \\ \frac{\partial G_{\log}}{\partial R_t} &= \frac{252}{T(1 + R_t)}. \end{aligned} \quad \text{(S37)}$$

On the variance-floor branch, the derivative of the floored denominator is zero. The execution graph propagates these sensitivities through returns, costs, positions, and continuous portfolio drift.

B. Two-pass gradient replay

Keeping every daily encoder and solver graph for a complete return path is memory-intensive. The replay uses the vector–Jacobian product of reverse-mode differentiation [31] and the activation-recomputation principle of gradient checkpointing [32]. Our split is at the sequence of daily neural targets, where the trading ledger supplies one adjoint per decision. The first pass computes and records the neural targets, discrete route and support choices, random state, and auxiliary sampling decisions. The ledger then treats target weights as differentiable leaves and obtains

$$\lambda_t = \frac{\partial L_{\mathrm{econ}}}{\partial w_t} \in \mathbb{R}^{N_t}. \quad \text{(S38)}$$

The adjoint includes effects propagated to subsequent ledger states, not just the immediate day's return. The second pass reconstructs one decision's neural graph at a time, under the same parameters and replayed decisions, and backpropagates

$$L_t^{\text{replay}} = \text{sg}(\lambda_t)^\top w_t(\theta) + T^{-1} L_{\text{aux},t}(\theta). \tag{S39}$$

The $1/T$ in the economic path objective is already contained in λ_t; its dot product is not divided by T again.

Proposition 6 (Conditional replay chain rule): For a fixed discrete execution and neural-computation branch, identical parameters and replayed randomness, summing the gradients of (S39) gives the continuous chain-rule gradient of (S36) for the implemented computation.

Proof: Condition on the recorded branch and write the differentiable targets as $w_t = f_t(\theta)$. The reverse-mode chain rule [31] on this fixed branch gives

$$\nabla_\theta L = \sum_t (\partial f_t/\partial\theta)^\top \lambda_t + T^{-1} \sum_t \nabla_\theta L_{\text{aux},t}. \tag{S40}$$

Stopping gradients through λ_t in the replay surrogate yields exactly its first summand, while the explicit auxiliary term yields the second. The execution adjoint already accounts for continuous dependencies between successive ledger states. ■

The conditioned branch includes discrete routing, admission and held masks, stopping, selected feasible candidate, and the implementation's detached feedback and targets. Replay does not differentiate these selection rules or convert them into a continuous relaxation. Forward agreement is checked before accepting replay gradients. The actual held mask is the temporal input to the formal neural forward call; it does not receive a continuous previous-weight vector. The separate ledger supplies continuous portfolio carry.

After accumulating one complete path gradient, AdamW [33] performs one update with zero weight decay, learning rate 10^{-4} for the bound parameter groups, and global gradient-norm clipping [34] at 5. Clipping occurs once after full-path accumulation, not separately per decision. The formal configuration does not introduce optional temporal-prior or extra side-cap branches.

TABLE S1
ARCHITECTURE AND NUMERICAL SETTINGS USED IN OUR REFERENCE IMPLEMENTATION.

Component	Bound configuration
Daily input	252 observations, 71 channels
Monthly input	Two 12-month blocks, 51 channels
Temporal encoders	One layer, width 256, four heads
Temporal FFN / dropout	1024 / 0.1
Shared summaries / keys	64 / 32 dimensions
Long-budget range	$\beta \in (0.02, 0.98)$
Diagnostic requests	32, equally divided by side
Source sparsity / fraction	Four stocks / 0.25
Refinement rounds	At most four per decision
Retrieval waves / leaf size	Two / at most 64 stocks
Access-value heads	Three; disagreement weight 0.25
Access-work proxy	$n_c + 4n_f + 32$
Coarse / node / fine budget	16,384 / 32,768 / 1,024
Exploration fraction	0.125 of access allocation
Plan versions / factors	At most 64 / eight dimensions
Gain and label scale κ	0.02
Quadratic coefficients	$\lambda_D = 0.05$, $\varepsilon_T = 10^{-6}$
Revision penalty λ_{rep}	0.0005 (internal objective only)
Base execution fee c_f	0.0005 per absolute executed weight
Forward ADMM	48 steps, $\rho = 0.5$
Solver convergence check	Infinity-norm tolerance 10^{-4}
Writeback coordinate bound	0.25
Return component L_{ret}	$-S_{\text{tr}} - 0.25 G_{\log}$
Optimizer / gradient clip	AdamW, norm 5 after path accumulation
Learning rate / weight decay	10^{-4} / zero

S9. Data Interfaces, Complexity, and Reproducibility

A. Point-in-time data contracts

CRSP supplies the trading and corporate-action records. Date-valid CRSP/Compustat Merged (CCM) links match Compustat accounts to CRSP securities [35]. The data interface binds ordered feature names, permanent stock identifiers, masks, reference-universe definitions, and availability timestamps to each sample. Daily feature transformation uses observed values in the same-date reference universe: winsorization [36] at the 1st and 99th percentiles, followed by the configured cross-sectional standardization. Winsorization limits extreme magnitudes while preserving the central 98% of observed values. The same U.S. reference universe is retained across evaluated subuniverses to keep their feature scales comparable. Transformation statistics never use subsequent dates. Monthly features follow their separately bound historical contract. Raw executable prices and corporate actions remain separate from normalized model inputs.

Shared quarterly shards and NumPy memory-mapped arrays [37] support lazy window assembly rather than materializing every overlapping 252-day tensor. The loader aligns both frequencies by permanent identifier and decision timestamp, checks finite values after imputation, and retains observation masks. The execution layer distinguishes a signal timestamp from an executable-price timestamp and keeps a consistent event order. Missing new-order execution, suspended held positions, and delisting settlements are ledger cases with explicit policies and diagnostics, not reasons to rewrite a decision-time feature tensor.

A reproducible run records the data vintage, common-universe policy, feature schema and order, normalization reference, eligible masks, execution and valuation conventions, cost coefficients, optimization configuration, and random states. Checkpoints bind these contracts to model and optimizer state. Split definitions and evaluation protocols are specified by the experiment, outside this methodology.

B. Computational profile

For fixed widths, the summary operator replaces dense cross-stock attention with $O(NC)$ stock-summary interactions; diagnostics require $O(NJ)$. Applying the standard self-attention cost [7] to each stock gives $O(NL_D^2 d)$ and

$O(N\,2L_M^2 d)$, plus projection and feed-forward work, where $L_D = 252$ and $L_M = 12$.

The median splits give $O(\log N)$ levels. Each level partitions at most N stocks. With an $O(n \log n)$ sort for a node of size n [38], the total sorting cost per level is at most $O(N \log N)$. Summing over levels yields $O(N \log^2 N)$ construction work. Storing one membership per stock per level requires $O(N \log N)$ space. Access budgets control the number of expensive inspected edges. Eligibility is represented over $N \times J$, while transfer directions and withdrawal capacities can be dense $N \times P$ matrices. For $P \leq 64$, the solver's matrix construction costs approximately $O(NP^2)$ and dense factorization $O(P^3)$ [20, App. C]. With the Cholesky factor reused, an ADMM step costs $O(NP + P^2)$, apart from additional linear rows. The number of simultaneous plan versions therefore directly controls these costs.

Replay bounds neural-activation storage to one decision's graph at a time. It also retains trajectory targets and execution states, typically $O(TN)$, together with data and route metadata.

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